

Algorithm for Locating Multiple Leaks in a Pipeline with an Invariance Condition^{*}

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Abstract: This paper addresses the problem of multiple leaks' location in a pipeline by considering available measurements of pressure and flow rate at the ends of the pipeline. New conditions are derived that allow the establishment of a well-posed identification problem for two leaks. These conditions are deduced from the existence of the redundant description of the intermediate flow in the line, the invariance of the conduit's physical parameters with respect to the flow's operating points, and the relationship that links the hydraulic gradients of the respective scenarios of one and two leaks. These joint conditions, which have not been established before, eliminate the assumption of sequential leaks reported in the literature and are the paper's main contribution. Moreover, a minimization algorithm considering these conditions is introduced here. Synthetic data demonstrate the proposed algorithm's effectiveness for five situations.

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1. INTRODUCTION

Leaks in pipelines and storage systems are significant concerns in the chemical industry and water distribution networks since the number of abnormal scenarios that can occur in actual installations is very diverse. The damage caused by the leaks includes economic losses, ecological consequences, and human injuries. Regarding automatic leak location methods, several works have been proposed over the last 30 years (Moubayed et al., 2021).

The variety of techniques ranges from simple physical inspection and acoustic methods to optical sensors and infrared thermography, which are complementary to the supervisory control and data acquisition (SCADA) system (Fang et al., 2019; Farmer, 2017). Some of the techniques require isolating and shutting down part of the system. From a practical point of view, there is an interest in applying advanced diagnostic procedures via software where only flow rate and pressure head sensors are available because the installations are frequently instrumented with these types of sensors. One of the most traditional automatic leak location methods is based on observer theory and mass balance with flow rate and load measurements at the ends of the duct (Verde and Torres, 2017; Lopezlena and Sadovnychiy, 2019).

On the other hand, because the pipeline variables are noisy and are digitalized with a low-frequency sample, methods associated with transient behavior can feasibly be applied only in specific cases and installations. For a complete overview of transient fault detection issues, one can consult Verde and Torres (2017).

Concerning the different fault scenarios, the single leak diagnosis is the most studied for both main lines and networks (Puig et al., 2017). On the contrary, the multi-leak problem is rarely studied, and most propositions have been tackled with linearized models in the frequency domain (Duan, 2018; Peralta and Verde, 2023; Rui et al., 2017). A disadvantage of the methods based on linearization is the sensitivity of the hydraulic gradient function to diverse uncertainties, such as friction and wave speed in the system. A significant result in the context of multi-leak isolation is reported in Verde et al. (2003). The authors showed that it is not feasible to isolate two simultaneous leaks if the fluid is in steady state and that only pressure and flow measurements are available at the ends of the pipeline. This weak condition is a challenge and motivates the search for additional conditions for the multi-leak localization.

A method to partially solve the multi-leak problem in real time was proposed in Rojas and Verde (2020) by assuming sequential leaks. By designing an iterative process, an extended Kalman filter estimates a single equivalent leak. In addition, from the equivalent model between one and multiple leaks, as well the process history, the parameters of the last leak are obtained. The key point of the method is the equivalence relation that links the hydraulic gradients of the scenarios of one and two leaks. Experimental data show the achievements of the scheme.

The aforementioned facts about the impossibility of parameter identification for two leaks motivated this contribution. Here new conditions for the two leaks' estimation are developed, so the assumptions of sequential leaks and of a linear model can be eliminated.

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The conditions are based on the invariance of orifices' physical parameters with respect to the flow's operation point and on the possibility of estimating an intermediate flow rate from pipeline's upstream and downstream data. An advantage of these features is that they allow proposing an error cost function J to be minimized in a set of flow operation points with constraints for the parameter identification of two leaks. The new conditions combined with the proposed J criterion are the main contributions of this work. Taking into account the formulation of the minimization task, here an algorithm that uses the gradient method is described. The algorithm offers the possibility of being implemented off-line or in real time.

The content of the paper is organized as follows. Section 2 introduces the input-output relationship for multi-leak scenarios in a hydraulic pipeline. In Section 3, invariance conditions for the two-leak isolation problem are presented. The implementation of the location-based algorithm is introduced in Section 4, while Section 5 presents the performance of the algorithm by means of simulation results that consider pipeline parameters taken from a real hydraulic pilot installation. Finally, Section 6 shows the conclusions.

2. INPUT-OUTPUT RELATIONSHIP FOR MULTI-LEAK SCENARIOS

Consider the dynamics of a pressurized incompressible fluid in a one-dimensional duct for a turbulence flow with constant density and convective velocity, which according to Chaudhry (2013) is described by the momentum and continuity equations

$$\frac{1}{gA} \frac{\partial q(z, t)}{\partial t} + \frac{\partial h(z, t)}{\partial z} + \theta q^2(z, t) = 0, \quad (1)$$

$$\frac{\partial h(z, t)}{\partial t} + \frac{a^2}{gA} \frac{\partial q(z, t)}{\partial z} = 0, \quad (2)$$

where $h(z, t)$ [m] and $q(z, t)$ [m³/s] denote the piezometric head and flow rate at the center of the pipe, respectively, $z \in [0, \ell]$ [m] is the spatial variable for a line of length ℓ , and $t \geq 0$ [s] is the time variable. The parameters of the line are as follows: the gravitational acceleration g [m/s²], the constant cross-sectional area A [m²], the pressure wave speed a [m/s], and constant $\theta = \frac{f}{2gDA^2}$ [s²/m⁶], which is associated with the system friction where D is the pipe diameter. Furthermore, f is the Darcy-Weissbach friction factor and depends on the Reynolds number, viscosity, and roughness (Torres and Verde, 2018; Xu and Karney, 2017). For the system (1)-(2), the piezometric heads $h(0, t)$ and $h(\ell, t)$ are assigned the boundary conditions.

Any leak with an outflow rate $q_f(t)$ at arbitrary position ℓ_f of the line produces a discontinuity in the model (1)-(2), and the piezometric head at ℓ_f induces the outflow rate

$$q_f(t) = \begin{cases} \lambda_f \sqrt{h(\ell_f, t)}, & \text{if } t \geq \tau \geq 0, \\ 0, & \text{o.w.,} \end{cases} \quad (3)$$

with τ as the leak occurrence time. The constant leak coefficient $\lambda_f > 0$ depends on the discharge coefficient, the leak's cross-sectional area, and the gravitational acceleration.

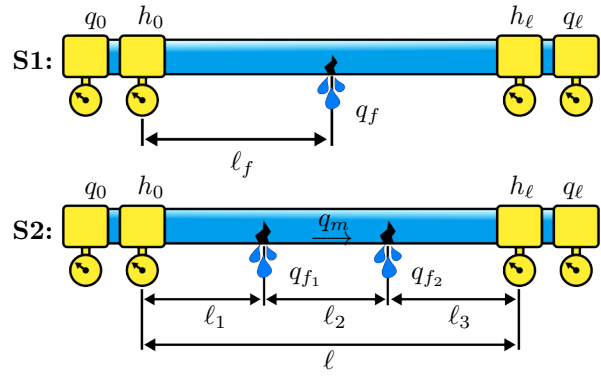


Fig. 1. The variables' description of a pipeline with scenarios S1 with a leak at ℓ_f and S2 with two leaks at ℓ_1 and $\ell_1 + \ell_2$, and measurement data $\mathcal{M} = \{h_0, h_\ell, q_0, q_\ell\}$

A pipeline with leaks can be modeled by a set of healthy line segments modeled by (1)-(2) and linked by boundary conditions associated with the leak junction by

$$\begin{bmatrix} h(\ell_f - \epsilon, t) \\ q(\ell_f - \epsilon, t) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} h(\ell_f + \epsilon, t) \\ q(\ell_f + \epsilon, t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} q_f(t) \quad (4)$$

for each position ℓ_f with $\epsilon \rightarrow 0$.

By considering the two scenarios shown in Fig.1, both scenarios can be modeled by healthy line segments connected with fault junctions. S1 scenario consists of an outflow $q_f(t)$ at position ℓ_f and the S2 scenario consists of outflows $q_{f1}(t)$ and $q_{f2}(t)$ at positions ℓ_1 and $\ell_1 + \ell_2$ with the middle flow rate q_m between the two leaks. Thus, the block diagrams shown in Fig. 2 model both scenarios. The S1 scenario is given by two blue blocks, which each represent a partial differential equations (PDE) set (1)-(2) joined by a red block that models a leak by (4). For the S2 scenario, the three blocks associated with the healthy sections are PDEs of the form (1)-(2) and are joined by the respective leak models of the form (4).

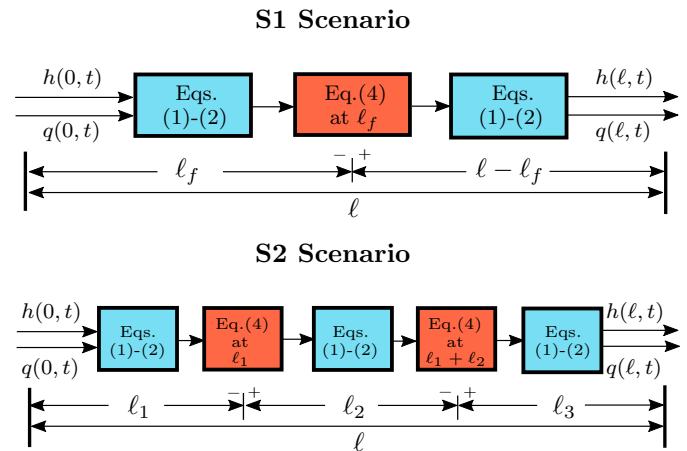


Fig. 2. Block diagram models for scenarios S1 and S2 and the same value set $\mathcal{M}$

If the pipeline state is in equilibrium, the parameter identification task for the S2 scenario is a challenge from a fault diagnosis point of view because the model (1)-(2) is reduced to

$$\frac{\partial h(z)}{\partial z} = \theta q^2(z), \quad \frac{\partial q(z)}{\partial z} = 0, \quad (5)$$

and this generates a lack of isolability for the outflows q_{f1} and q_{f2} . By assuming there are sequential leaks in the S2 scenario, the problem is partially solved. Here one eliminates the condition of sequential occurrence by adding a physical condition to the data that involves a simple, feasible maneuver in the pipeline. The solution proposed is based on the fact that (ℓ_i, λ_i) are parameters associated with the geometry and physics of the orifices, and, therefore, they are invariant for any value set at the ends of line $\mathcal{M}$. The new isolability conditions are presented below.

2.1 Input-Output Equivalent Relationship

Consider the invariant leak parameter sets $\mathcal{P}_1 = \{\ell_f, \lambda_f\}$, $\mathcal{P}_2 = \{\ell_1, \ell_2, \lambda_1, \lambda_2\}$ for the two scenarios shown in Fig. 2 where the two-port blocks correspond with the spatially discretized models (4) and (5) and with available data set $\mathcal{M}$. Thus, from the mass balance for each scenario, one gets the outflow

$$q_f = q_0 - q_\ell = \begin{cases} \lambda_f \sqrt{h_0 - \ell_f \theta q_0^2} & \text{for S1} \\ q_{f1} + q_{f2} & \text{for S2,} \end{cases} \quad (6)$$

with

$$q_{f1} = \lambda_1 \sqrt{h_0 - \ell_1 \theta q_0^2}, \quad (7)$$

$$q_{f2} = \lambda_2 \sqrt{h_\ell + (\ell - \ell_1 - \ell_2) \theta q_\ell^2}, \quad (8)$$

after some algebraic manipulations (Verde and Torres, 2017). If ℓ_1 and ℓ_2 are known, one can note from (6) to (8) that λ_1 and λ_2 are characterized by an affine function. Moreover, if the value set $\mathcal{M}$ for both scenarios is assumed equal, the positions in S1 and S2 are related to the nonlinear function

$$\ell_f = \frac{h_0 - h_\ell - \ell \theta q_\ell^2}{\theta (q_0^2 - q_\ell^2)} = \ell_1 + \ell_2 \left(\frac{q_m^2 - q_\ell^2}{q_0^2 - q_\ell^2} \right). \quad (9)$$

This association between scenarios is denoted as an equivalent relation (ER), and the leaks' positions satisfy the inequality

$$0 < \ell_1 < \ell_f < \ell_1 + \ell_2 < \ell. \quad (10)$$

An important property of the flow rate of the intermediate section q_m involved in (9) is that it can be expressed in terms of upstream or downstream variables such that

$$q_m = \begin{cases} q_0 - q_{f1} & \text{for upstream data} \\ q_\ell + q_{f2} & \text{for downstream data} \end{cases} \quad (11)$$

is satisfied. Thus, one can take advantage of this redundant description to indistinctly estimate q_m by using upstream or downstream data.

2.2 $\mathcal{P}_2$ Identifiability Analysis through the ER

To analyze the implications of the conditions given by the ER and the redundant description of q_m in terms of hydraulic gradient functions per unit length of the line, the profile of the piezometric pressure head in the line is developed.

Since m_0 , m_1 and m_3 are known, by considering the system

$$\begin{pmatrix} m_0 \\ m_1 \\ m_2 \\ m_3 \end{pmatrix} = \theta \begin{pmatrix} q_v^2 \\ q_0^2 \\ q_m^2 \\ q_\ell^2 \end{pmatrix} = \begin{pmatrix} \ell^{-1} & 0 & 0 & -\ell^{-1} \\ \ell_1^{-1} & -\ell_1^{-1} & 0 & 0 \\ 0 & \ell_2^{-1} & -\ell_2^{-1} & 0 \\ 0 & 0 & \ell_3^{-1} & -\ell_3^{-1} \end{pmatrix} \begin{pmatrix} h_0 \\ h_{f1} \\ h_{f2} \\ h_\ell \end{pmatrix} \quad (12)$$

with q_v as the virtual flow rate without leaks in the line, one obtains

$$\ell_f = \ell \left(\frac{m_0 - m_3}{m_1 - m_3} \right) = \ell_1 + \ell_2 \left(\frac{m_2 - m_3}{m_1 - m_3} \right), \quad (13)$$

which is equivalent to (9) where the only unknown gradient function is m_2 and can be estimated using

$$m_2 = \begin{cases} m_2(\ell_1, \lambda_1) = \theta \left(q_0 - \lambda_1 \sqrt{h_0 - \ell_1 \theta q_0^2} \right)^2, \\ m_2(\ell_1, \ell_2, \lambda_2) = \theta \left(q_\ell + \lambda_2 \sqrt{h_\ell + (\ell - \ell_1 - \ell_2) \theta q_\ell^2} \right)^2. \end{cases} \quad (14)$$

Regarding m_2 , there is again a redundancy in its estimation, which is similar to the flow rate (11). The first expression is only a function of the upstream measurements of the middle section, and the second is only a function of the downstream measurements.

The resulting piezometric pressure profile by using (13) is the concatenation of the lines $A'A$, AB , and BB' shown in Fig. 3. This graphic confirms the impossibility of identifying the parameters of the triangle ABC if only slopes m_1 and m_3 and vertex C are assumed to be known.

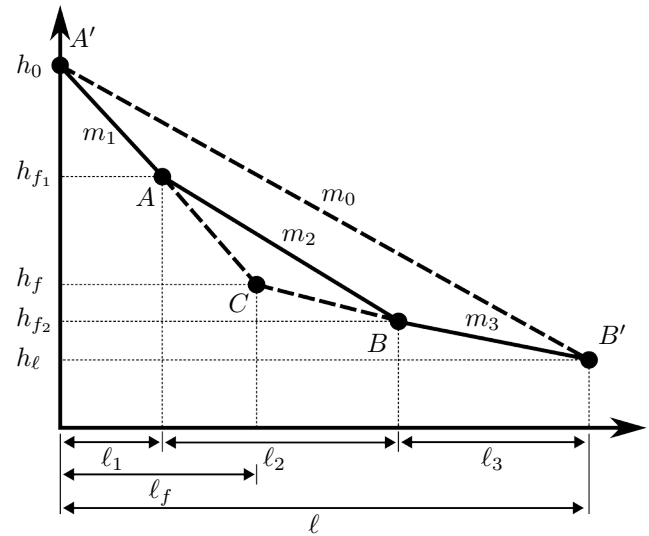


Fig. 3. Piezometric pressure profile for two leaks in the line with unknown slope m_2

3. CONDITIONS FOR TWO LEAKS' ISOLATION

The following conditions that allow formulating the identifiability problem for the parameter set $\mathcal{P}_2$ can be extracted from the pressure profile analysis, the redundancy of the hydraulic gradient m_2 , and the understanding of the virtual position ℓ_f .

C1: The physical parameters $\ell_1|_i$ and $\ell_2|_i$ possess an invariance with respect to the steady operation point and satisfy the inequality

$$0 < \ell_1 < \ell_1 + \ell_2 < \ell.$$

- C2: According to Verde et al. (2016), the virtual position $\ell_f|_i$ is sensitive to any operating point and corresponds with the ordinate of the vertex C in the Fig. 3.
- C3: The virtual position $\ell_f|_i$ can be estimated in terms of upstream or downstream data for the operation point i by

$$\hat{\ell}_f|_i = \begin{cases} \hat{\ell}_f^u|_i \triangleq \ell_1 + \ell_2 \left(\frac{m_2(\ell_1, \lambda_1) - m_3}{m_1 - m_3} \right) \Big|_i, \\ \hat{\ell}_f^d|_i \triangleq \ell_1 + \ell_2 \left(\frac{m_2(\ell_1, \ell_2, \lambda_2) - m_3}{m_1 - m_3} \right) \Big|_i, \end{cases} \quad (15)$$

acquiring the redundancy of m_2 .

- C4: The virtual position $\ell_f|_i$ for each operation point i can also be calculated with known data from $\mathcal{M}|_i$ by

$$\ell_f|_i = \ell \left(\frac{m_0 - m_3}{m_1 - m_3} \right) \Big|_i. \quad (16)$$

- C5: The physical parameters $\lambda_1|_i$ and $\lambda_2|_i$ associated with the outflow rate possess an invariance with respect to the operation point and satisfy the non-linear functions

$$\lambda_1|_i = \left(\frac{q_0 - \sqrt{\frac{\ell m_0 - \ell_1 m_1 - (\ell - \ell_1 - \ell_2) m_3}{\theta \ell_2}}}{\sqrt{h_0 - \ell_1 m_1}} \right) \Big|_i, \quad (17)$$

$$\lambda_2|_i = \left(\frac{q_0 - q_\ell - \lambda_1 \sqrt{h_0 - \ell_1 m_1}}{\sqrt{h_\ell + (\ell - \ell_1 - \ell_2) m_3}} \right) \Big|_i. \quad (18)$$

Note that functions (17) and (18) prove the impossibility of identifying the set $\mathcal{P}_2$ with four unknown parameters if a single operation point is considered. Nevertheless, if a set of operation conditions is available, one can build a surface such that given ℓ_1 one can determine ℓ_2 from (17) and (18).

Since all the conditions given above are necessary, one can establish the identification problem as the search for set $\mathcal{P}_2$ such that the errors of the virtual position given by

$$e^u|_i = \ell_f|_i - \hat{\ell}_f^u|_i \quad \forall i \quad (19)$$

with estimated upstream data and

$$e^d|_i = \ell_f|_i - \hat{\ell}_f^d|_i \quad \forall i \quad (20)$$

with estimated downstream data both become 0. This formulation implicitly satisfies the conditions C2, C3, and C4. With respect to C1, since ℓ_1 and ℓ_2 are assumed constant in the above derivation, if the search domains for the parameters ℓ_1 and ℓ_2 satisfy the inequality

$$0 < \ell_1 < \ell_f|_i < \ell_1 + \ell_2 < \ell \quad \forall i, \quad (21)$$

C1 is accomplished.

By considering the invariance condition C5 of the orifice parameter λ_1 , this condition is equivalent to the surface

$$\left(\frac{q_0 \sqrt{\theta \ell_2} - \sqrt{\ell m_0 - \ell_1 m_1 - (\ell - \ell_1 - \ell_2) m_3}}{\sqrt{h_0 - \ell_1 m_1}} \right) \Big|_1 = \left(\frac{q_0 \sqrt{\theta \ell_2} - \sqrt{\ell m_0 - \ell_1 m_1 - (\ell - \ell_1 - \ell_2) m_3}}{\sqrt{h_0 - \ell_1 m_1}} \right) \Big|_2. \quad (22)$$

Thus, if one applies a digital solver for (22), the value set of ℓ_2 that satisfies C5 can be obtained for any ℓ_1 .

Thus, here one proposes the error cost function

$$J(e^u, e^d) = \frac{1}{n} \sum_{i=1}^n (e^u|_i^2 + e^d|_i^2) \quad (23)$$

for the estimation of $\mathcal{P}_2$ where the inequality (21) and the invariance condition (22) must be satisfied with $n > 2$, which is the number of available measured in steady-state $\mathcal{M}|_i$.

Because in an absence of leaks the hydraulic gradient function characterized by m_0 , and ℓ_f is undetermined, one assumes the existence of an active leak alarm for the isolation task. Considering that the residual generator and the alarm signal are well-discussed topics by the Safe Process Community, they are omitted here for space reasons. The reader can consult the classic references for these subjects (Isermann, 1997; Blanke et al., 2006; Korbicz et al., 2004).

Consequently, the multi-leak location problem can be formulated as the follow minimization task:

$$\begin{aligned} J^*(e^u, e^d) = \min_{\mathcal{P}_2} J(e^u|_i, e^d|_i), \quad \forall i \\ \text{subject to} \\ \text{the invariance of } \lambda_1 \text{ and } \lambda_2, \\ 0 < \ell_1 < \ell_f|_i < \ell_1 + \ell_2 < \ell, \end{aligned} \quad (24)$$

and an activated alarm condition, regardless of the leaks' number. This idea can be extended in the case of n leaks in the line increasing the number of flow's operating points.

4. TWO-LEAK LOCATING ALGORITHM

Since the proposed error function $J(e^u, e^d)$ and its gradient with respect to ℓ_1 are continuous (as seen in Appendix A), to achieve

$$\mathcal{P}_2^* = \arg \min J(e^u, e^d), \quad (25)$$

the descending sequence

$$\ell_1(k+1) = \ell_1(k) - \alpha \frac{\partial J(k)}{\partial \ell_1(k)} \quad (26)$$

is applied with $\ell_f > \ell_1 > 0$, where the step size or learning rate α establishes a compromise between accuracy and efficiency (Luenberger and Ye, 2015). An advantage of the cost function and this iterative form for adjusting ℓ_1 is that the search domain is limited to the interval $[0, \ell_f]$ and that the gradient $\frac{\partial J}{\partial \ell_1}$ is 0 only if both estimators ℓ_f^u and ℓ_f^d converge at the equivalent position using upstream and downstream data, respectively. This is a robust property of the procedure.

From an application point of view, two implementations can be designed for the minimization task: one assuming a quasi-real-time and the other off-line. This section outlines a specific off-line algorithm by considering only three available operation points for a hydraulic pipeline.

5. SIMULATION RESULTS

The performance of the algorithm is analyzed with synthetic data from the hydraulic pilot pipeline with parameters that are given in Table 1 (Rojas and Verde, 2020). Five different leak cases are represented in the study and are found in Table 2. For each case, three operating points are

Algorithm 1 Two-leak Locating Algorithm by $q_0 \neq q_\ell$

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1: Input data:  $\mathcal{M}|_i$  for  $i = 1, \dots, n$ 
2: Output:  $\mathcal{P}_2^* = \{\ell_1^*, \ell_2^*, \lambda_1^*, \lambda_2^*\}$ 
3: Set convergence criterion:  $\epsilon_0 > 0$ 
4: Set the step size:  $\alpha > 0$ 
5: if the alarm is active then
6:   Compute  $\ell_f|_i$ , by (16) for  $i = 1, \dots, n$ 
7:   Initial parameters:  $k = 1, \ell_1(1) = 0$ 
8:   do
9:     Estimate the set  $\bar{L}_2$  from (22) with  $\ell_1(k)$ 
10:    Select  $\ell_2(k)$  according to (21) by
11:    for  $j = 1 : |\bar{L}_2|$  do
12:      if  $\ell_1(k)$  and  $L_2(j)$  satisfy (21) then
13:         $\ell_2(k) = L_2(j)$ 
14:      else Warning: no solution. Go to end while
15:    end if
16:  end for
17:  Compute  $\lambda_1(k)$  and  $\lambda_2(k)$  with (17) and (18)
18:  Estimate  $\hat{\ell}_f^u(k)|_i$  and  $\hat{\ell}_f^d(k)|_i$  for  $i = 1, \dots, n$ 
    with (15)
19:  Evaluate  $J(k)$  with (23)
20:  Evaluate  $\Delta J(k) = \frac{\partial J}{\partial \ell_1}$  with (A.1)
21:  Update  $\ell_1(k+1) = \ell_1(k) - \alpha * \Delta J(k)$ 
22:   $k = k + 1$ 
23:  while  $|\Delta J| > \epsilon_0$ 
24: end if
25: Stop

```

simulated, which generate a database with a time window of 100 s. From this database, the average values of $\mathcal{M}|_i$ for $i = 1, 2$, and 3 are calculated and become the input data for the algorithm.

Table 1. Pilot pipeline parameters at UNAM

Parameter meaning	Symbols [Units]	Values
Length	ℓ [m]	160.76
Pipe diameter	D [m]	0.076
Gravity acceleration	g [m/s ²]	9.81
Speed wave	a [m/s]	1560
D-W friction factor	f	0.0179
	θ [s ² /m ⁶]	583.32

Table 2. Leak parameters for the five analyzed situations

Case	Parameters			
	ℓ_1 [m]	ℓ_2 [m]	λ_1 [m ^{5/2} /s]	λ_2 [m ^{5/2} /s]
A	5.97	148.85	1.60×10^{-4}	2.30×10^{-4}
B	41.05	78.89	2.20×10^{-4}	1.15×10^{-4}
C	63.01	34.88	2.09×10^{-4}	2.09×10^{-4}
D	5.97	57.04	1.50×10^{-4}	1.00×10^{-4}
E	97.89	56.93	2.10×10^{-4}	1.10×10^{-4}

In this particular study, $\epsilon_0 = 1 \times 10^{-5}$ and $\alpha = 500$ are chosen. It is recommended to start at ℓ_1 close to its boundaries $[0 - \ell_f]$. Here $\ell_1 = 0$ is used. Thus, under these conditions, the implementation of the algorithm achieves the estimated parameters given in Table 3 where the respective deviations are shown. The deviations $\delta \mathcal{P}_2^*$ show a satisfactory estimation with the worst relative error in the order of 12.90% in the estimation of ℓ_1 for case D. One remarks that even the leaks that are very near the ends of the line are properly estimated (case A).

Table 3. Leak parameters and results with the relative deviation in percentage $\delta \mathcal{P}_2^*$

Case		Parameters			
		ℓ_1^* [m]	ℓ_2^* [m]	λ_1^* [m ^{5/2} /s]	λ_2^* [m ^{5/2} /s]
A	$\mathcal{P}_2^*$ $\delta \mathcal{P}_2^*$	5.85 2.01%	148.95 0.07%	1.60×10^{-4} 0%	2.30×10^{-4} 0%
B	$\mathcal{P}_2^*$ $\delta \mathcal{P}_2^*$	40.18 2.12%	78.45 0.56%	2.15×10^{-4} 2.27%	1.20×10^{-4} 4.35%
C	$\mathcal{P}_2^*$ $\delta \mathcal{P}_2^*$	62.16 1.35%	34.70 0.52%	1.98×10^{-4} 5.26%	2.20×10^{-4} 5.26%
D	$\mathcal{P}_2^*$ $\delta \mathcal{P}_2^*$	5.20 12.90%	56.85 0.33%	1.46×10^{-4} 2.67%	1.04×10^{-4} 4.00%
E	$\mathcal{P}_2^*$ $\delta \mathcal{P}_2^*$	98.52 0.64%	57.11 0.32%	2.14×10^{-4} 1.90%	1.06×10^{-4} 3.64%

For a detailed presentation of the algorithm convergence process and advantages, only case E is presented for space reasons. Table 4 shows the average values of $\mathcal{M}|_i$ for $i = 1, 2, 3$. Figure 4 shows the evolution of the criterion $J(k)$ as a function of the sequence $\ell_1(k)$, where the value $J^* = 8.26 \times 10^{-3}$ is achieved after 35 iterations. Figure 5 illustrates the nonlinear relationship between the pair $(\ell_1(k), \ell_2(k))$ that holds the constraints.

Finally, Fig. 6 validates the affinity between $\lambda_1(k)$ and $\lambda_2(k)$. Thus, when the parameters $\mathcal{P}_2(k)$ for any two iterations are known, the rest of the pairs (λ_1, λ_2) can be straightforwardly estimated by the affine function of Fig. 6 or by the solution of (17) and (18). This feature of the relationship of the orifice parameters simplifies their estimation in ideal cases.

Table 4. Average synthetic data of case E

Operating Point	h_0 [m]	h_ℓ [m]	q_0 [l/s]	q_ℓ [l/s]
1	21.27	3.65	13.97	13.08
2	21.77	3.66	14.16	13.26
3	22.26	3.66	14.35	13.45

6. CONCLUSIONS

This work proposes a solution for the identification task of multiple leaks in a pipeline. In particular, a set of necessary conditions to formulate a well-posed locating problem has been established. This implies the knowledge of the states at the ends of the line in at least two flow's operating points. Under these considerations, the minimization of a cost function with constraints was established, and the respective algorithm was developed. The test of the procedure with synthetic data for five leak situations generated estimation errors less than 13%. The validation of the procedure with experimental data of a hydraulic pilot pipeline is under development. Regarding the extension for the case on n leaks in the line, the computational cost is the main issue for a real-time application since the isolability condition requires data to be available at more operating points.

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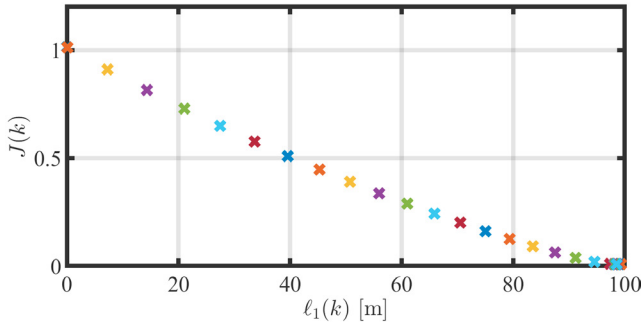


Fig. 4. Case E: Evolution of $J(k)$ as a function of $\ell_1(k)$ for each iteration.

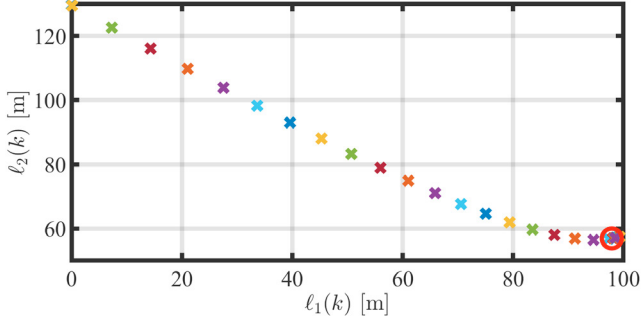


Fig. 5. Case E: Behavior of $\ell_1(k)$ and $\ell_2(k)$ for each iteration. The red circle indicates the positions of the real parameters.

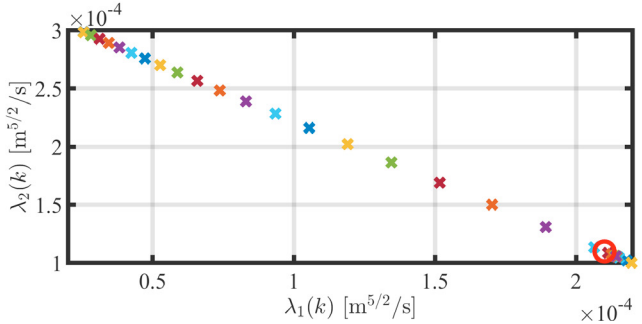


Fig. 6. Case E: Affinity of $\lambda_1(k)$ and $\lambda_2(k)$. The red circle indicates the orifices of the real parameters.

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Appendix A. GRADIENT FUNCTION

Gradient of the function $J(e^u, e^d)$ (23)

$$\frac{\partial J}{\partial \ell_1} = -\frac{2}{n} \sum_{i=1}^n \left((\ell_f|_i - \hat{\ell}_f^u|_i) \frac{\partial \hat{\ell}_f^u}{\partial \ell_1} \Big|_i + (\ell_f|_i - \hat{\ell}_f^d|_i) \frac{\partial \hat{\ell}_f^d}{\partial \ell_1} \Big|_i \right), \quad (\text{A.1})$$

with

$$\frac{\partial \hat{\ell}_f^u}{\partial \ell_1} \Big|_i = \left(1 + \frac{\ell_2 \lambda_1 \theta m_1}{m_1 - m_3} \left(\frac{q_0 - \lambda_1 \sqrt{h_0 - \ell_1 m_1}}{\sqrt{h_0 - \ell_1 m_1}} \right) \right) \Big|_i, \quad (\text{A.2})$$

$$\frac{\partial \hat{\ell}_f^d}{\partial \ell_1} \Big|_i = \left(1 - \frac{\ell_2 \lambda_2 \theta m_3}{m_1 - m_3} \left(\frac{q_\ell + \lambda_2 \sqrt{h_\ell + \ell_3 m_3}}{\sqrt{h_\ell + \ell_3 m_3}} \right) \right) \Big|_i. \quad (\text{A.3})$$